

Data Analysis & Probability



Mean, median, mode and range

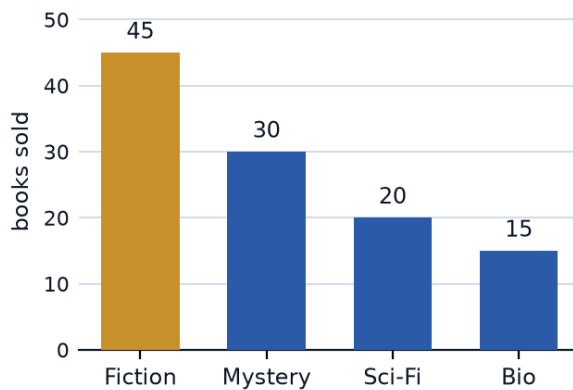
- 1 Find the mean of 6, 9, 12, 15, 18.
 - 2 Ages of 7 students: 13, 15, 13, 16, 14, 13, 17. Find the mean, median, and mode.
 - 3 Find the range and mode of the data set 8, 3, 8, 15, 8, 22, 3.
 - 4 A data set has mean 18 and 6 values. Find the sum of the values.
 - 5 The data set 10, 12, 11, 13, 50 contains an outlier. Find the mean before and after removing 50.
 - 6 A data set of 5 values has mean 20. Four of the values are 15, 18, 22, 25. Find the fifth value.
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Reading tables

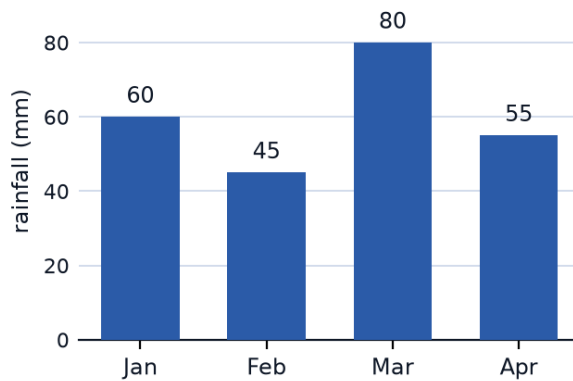
- 7 A survey of 50 people's favorite fruit: Apple 15, Banana 12, Orange 10, Mango 13. What fraction of people chose banana?
 - 8 Using the same survey (Apple 15, Banana 12, Orange 10, Mango 13, out of 50), what percent chose either apple or mango?
 - 9 A table shows quiz scores: score 5 (3 students), score 7 (5), score 8 (8), score 9 (4), score 10 (2). Find the mode score and the total number of students.
 - 10 Using the same quiz data, find the mean score to two decimal places.
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Reading bar charts

- 11 The bar chart shows books sold by genre in one week. Find the total books sold and the modal (most common) genre.



- 12 Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?
- 13 The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.



- 14 Using the same rainfall data (Jan 60, Feb 45, Mar 80, Apr 55 mm), find the percent decrease from March to April.

Basic probability

- 15 A bag holds 4 red and 6 blue marbles. Find $P(\text{red})$.
- 16 A fair six-sided die is rolled. Find $P(\text{a number greater than 4})$.
- 17 A card is drawn from a standard deck of 52 cards. Find $P(\text{a heart})$.
- 18 A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{an even number})$.
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Combined and complementary events

- 19 A bag holds 3 red, 5 blue and 2 green marbles. Find $P(\text{not red})$.
- 20 Two fair coins are tossed. Find $P(\text{both land heads})$.
- 21 A die is rolled twice. Find $P(\text{both rolls show a 6})$.
- 22 A bag holds 4 white and 6 black balls. One ball is drawn and not replaced, then a second is drawn. Find $P(\text{both white})$.
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Weighted averages

- 23 A class of 9 students has an average score of 80. A 10th student joins with a score of 100. Find the new average for all 10 students.
- 24 Class A has 20 students with mean score 75; Class B has 30 students with mean score 85. Find the overall mean of all 50 students.
- 25 A student's final grade is 40% homework average (score 90) and 60% exam average (score 70). Find the weighted final grade.
- 26 A survey combines two groups: 15 people with average age 24, and 25 people with average age 32. Find the overall average age of all 40 people.
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Comparing data sets

- 27 Data set A: 10, 12, 14, 16, 18. Data set B: 8, 14, 14, 14, 20. Both have the same mean. Which has the greater range?
- 28 Team X scored 22, 25, 28, 25 points in 4 games. Team Y scored 20, 20, 35, 25. Both teams have the same mean score. Which team's scores are more consistent (smaller range)?
- 29 Class 1 has 18 students with mean score 68; Class 2 has 22 students with mean score 74. Find the combined mean of all 40 students, to one decimal place.
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Word problems

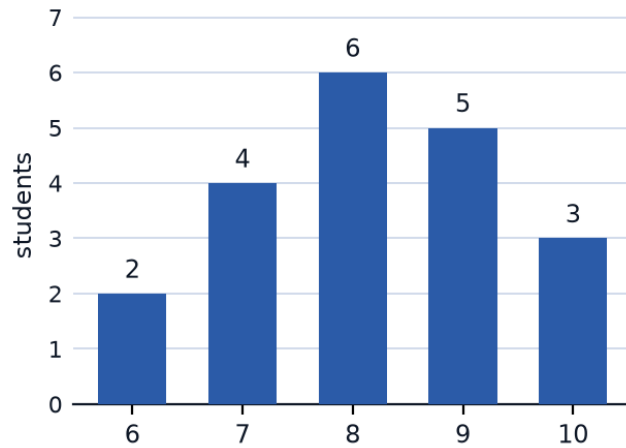
- 30 A teacher records the number of siblings for 10 students: 1, 2, 0, 1, 3, 2, 1, 0, 2, 1. Find the mean, median, and mode.
- 31 A company's weekly sales (in thousands of riyals) over 5 weeks: 40, 55, 35, 60, 50. Find the mean weekly sales, and state how many weeks had sales above the mean.
- 32 A jar contains 5 red, 3 blue and 2 yellow candies. A candy is picked and eaten, then another is picked. Find $P(\text{first red, then blue})$.
- 33 In a survey of 200 students, 120 prefer math and the rest prefer science. Find the probability, as a percent, that a randomly chosen student prefers science.
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MCQ — Mean, median, mode and range

- 34 Find the mean of 8, 11, 14, 17, 20.
- a. 12
 - b. 17
 - c. 70
 - d. 14
- 35 Ages of 6 students: 12, 14, 12, 15, 13, 12. Find the mean, median, and mode.
- a. Mean = 13, median = 12, mode = 12.5
 - b. Mean = 13, median = 12.5, mode = 12
 - c. Mean = 12.5, median = 13, mode = 12
 - d. Mean = 13, median = 12.5, mode = 13
- 36 Find the range and mode of the data set 5, 9, 5, 18, 5, 27, 9.
- a. Range = 13, mode = 5
 - b. Range = 27, mode = 5
 - c. Range = 22, mode = 5
 - d. Range = 22, mode = 9

- 37** A data set has mean 24 and 8 values. Find the sum of the values.
- a. 32
 - b. 3
 - c. 192
 - d. 96
- 38** The data set 14, 16, 15, 17, 60 contains an outlier. Find the mean before and after removing 60.
- a. Before = 122, after = 15.5
 - b. Before = 24.4, after = 15.5
 - c. Before = 15.5, after = 24.4
 - d. Before = 24.4, after = 62
- 39** A data set of 6 values has mean 25. Five of the values are 20, 22, 28, 30, 32. Find the sixth value.
- a. 18
 - b. 28
 - c. 8
 - d. 150

- 40 The bar chart shows quiz scores. Find the total number of students, and the modal score.



- a. 20 students; mode = 8
- b. 20 students; mode = 6
- c. 20 students; mode = 9
- d. 18 students; mode = 8

MCQ — Reading tables

- 41 A survey of 60 people's favorite drink: Tea 18, Coffee 22, Juice 12, Water 8. What fraction of people chose coffee?

- a. $\frac{18}{60}$
- b. $\frac{22}{60}$
- c. $\frac{11}{30}$
- d. $\frac{11}{60}$

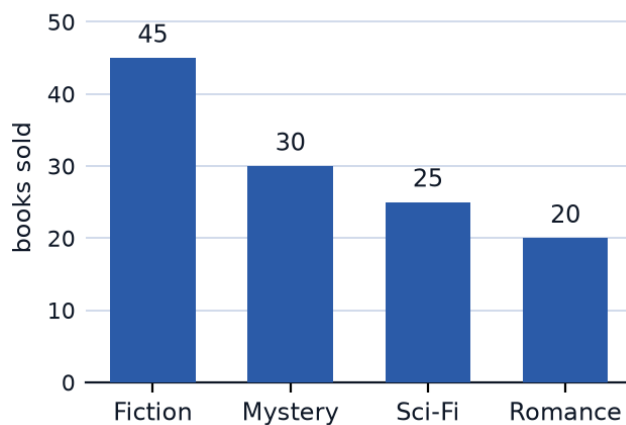
- 42 Using the same survey (Tea 18, Coffee 22, Juice 12, Water 8, out of 60), what percent chose either tea or water, to the nearest percent?
- a. 43%
 - b. 30%
 - c. 50%
 - d. 26%
- 43 A table shows test scores: score 60 (4 students), 70 (7), 80 (9), 90 (6), 100 (4). Find the total number of students.
- a. 34
 - b. 30
 - c. 26
 - d. 25
- 44 Using the same test data, find the mean score to two decimal places.
- a. 79.67
 - b. 80.00
 - c. 76.33
 - d. 79.00
- 45 A table shows the number of pets owned by 25 families: 0 pets (6 families), 1 pet (10), 2 pets (7), 3 pets (2). Find the median number of pets.
- a. 0
 - b. 1.2
 - c. 2
 - d. 1

46 Using the same pet data, find the mean number of pets per family, to two decimal places.

- a. 1.00
- b. 1.20
- c. 1.50
- d. 30

MCQ — Reading bar charts

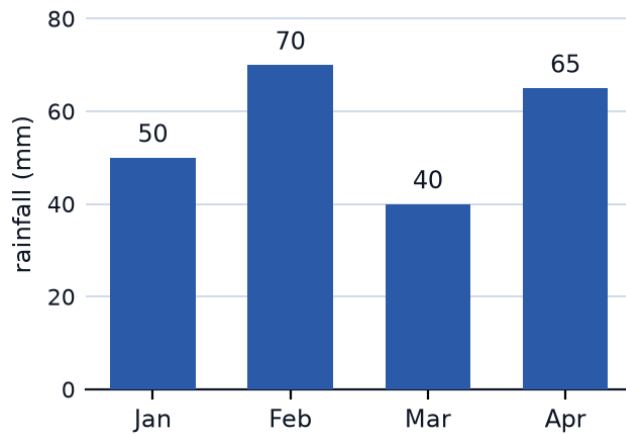
47 The bar chart shows books sold by genre in one week. Find the total books sold, and the modal genre.



- a. 120; Romance
- b. 100; Fiction
- c. 120; Mystery
- d. 120; Fiction

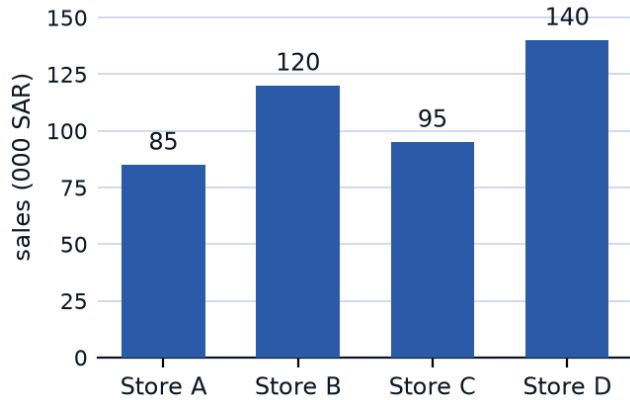
- 48 Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?
- a. 25%
 - b. 30%
 - c. 20%
 - d. 38%

- 49 The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.

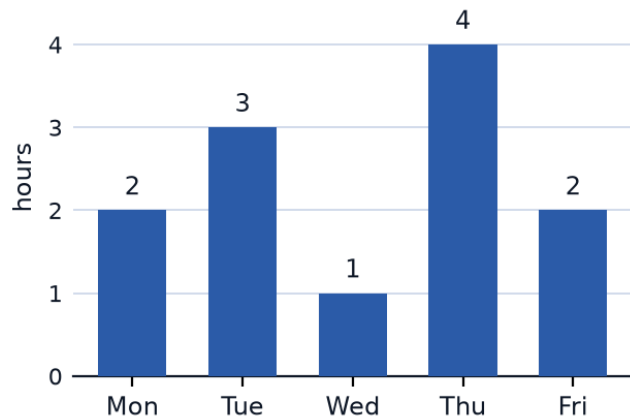


- a. 225
 - b. 65
 - c. 55
 - d. 56.25
- 50 Using the same rainfall data (Jan 50, Feb 70, Mar 40, Apr 65 mm), find the percent decrease from Feb to Mar, to the nearest percent.
- a. 75%
 - b. 43%
 - c. 30%
 - d. 40%

- 51 The bar chart shows quarterly sales for four stores, in thousands of riyals. Find the range, and which store had the highest sales.

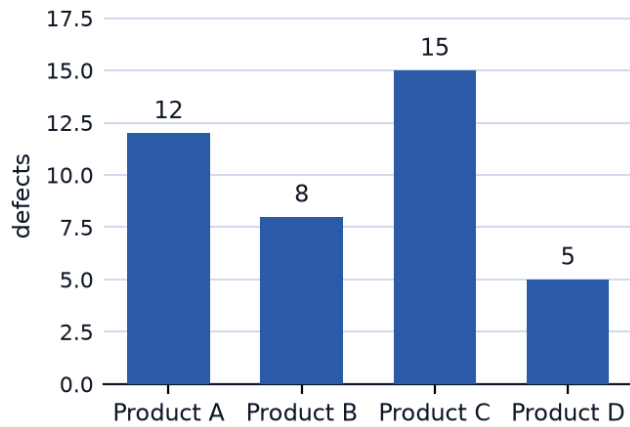


- a. 55; Store B
 - b. 55; Store D
 - c. 140; Store D
 - d. 35; Store D
- 52 The bar chart shows hours studied each weekday. Find the total hours, and the mean hours per day.



- a. 12; 2
- b. 12; 2.4
- c. 12; 3
- d. 10; 2

- 53 The bar chart shows the number of defective items found for four products in a quality check. Find the total defects, and the range.



- a. 40; 10
- b. 40; 7
- c. 35; 10
- d. 40; 15

MCQ — Basic probability

- 54 A bag holds 5 red and 7 blue marbles. Find $P(\text{red})$.

- a. $\frac{7}{12}$
- b. $\frac{5}{12}$
- c. $\frac{5}{7}$
- d. $\frac{1}{12}$

- 55 A fair six-sided die is rolled. Find $P(\text{a number less than 3})$.

- a. $\frac{1}{6}$
- b. $\frac{1}{2}$
- c. $\frac{2}{3}$
- d. $\frac{1}{3}$

56 A card is drawn from a standard deck of 52 cards. Find $P(\text{a king})$.

- a. $\frac{1}{4}$
- b. $\frac{1}{52}$
- c. $\frac{1}{13}$
- d. $\frac{4}{13}$

57 A spinner has 10 equal sections numbered 1 to 10. Find $P(\text{a multiple of 3})$.

- a. $\frac{1}{10}$
- b. $\frac{1}{3}$
- c. $\frac{4}{10}$
- d. $\frac{3}{10}$

58 A bag holds 6 yellow and 9 green balls. Find $P(\text{yellow})$.

- a. $\frac{1}{5}$
- b. $\frac{6}{9}$
- c. $\frac{3}{5}$
- d. $\frac{2}{5}$

59 A fair six-sided die is rolled. Find $P(\text{an odd number})$.

- a. $\frac{2}{3}$
- b. $\frac{1}{6}$
- c. $\frac{1}{2}$
- d. $\frac{1}{3}$

MCQ — Combined and complementary events

60 A bag holds 4 red, 6 blue and 5 green marbles. Find $P(\text{not blue})$.

- a. $\frac{3}{5}$
- b. $\frac{9}{6}$
- c. $\frac{2}{5}$
- d. $\frac{6}{15}$

61 Two fair coins are tossed. Find $P(\text{at least one tail})$.

- a. $\frac{3}{4}$
- b. $\frac{1}{3}$
- c. $\frac{1}{4}$
- d. $\frac{1}{2}$

62 A die is rolled twice. Find $P(\text{both rolls show an even number})$.

- a. $\frac{1}{6}$
- b. $\frac{1}{3}$
- c. $\frac{1}{4}$
- d. $\frac{1}{2}$

63 A bag holds 5 white and 7 black balls. One ball is drawn and not replaced, then a second is drawn. Find $P(\text{both black})$.

- a. $\frac{7}{22}$
- b. $\frac{6}{11}$
- c. $\frac{49}{144}$
- d. $\frac{7}{12}$

- 64 A card is drawn, replaced, then a second card is drawn from a standard deck. Find $P(\text{both hearts})$.
- a. $\frac{1}{8}$
 - b. $\frac{1}{2}$
 - c. $\frac{1}{4}$
 - d. $\frac{1}{16}$
- 65 A bag holds 3 red and 4 blue marbles. Two are drawn without replacement. Find $P(\text{both red})$.
- a. $\frac{3}{7}$
 - b. $\frac{9}{49}$
 - c. $\frac{2}{7}$
 - d. $\frac{1}{7}$
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MCQ — Weighted averages

- 66 A class of 12 students has an average score of 78. A 13th student joins with a score of 91. Find the new average.
- a. 84.50
 - b. 78.00
 - c. 91.00
 - d. 79.00
- 67 Class A has 25 students with mean score 70; Class B has 15 students with mean score 90. Find the combined mean.
- a. 80
 - b. 75
 - c. 160
 - d. 77.5

- 68** A student's final grade is 30% homework average (score 85) and 70% exam average (score 75). Find the final grade.
- a. 82
 - b. 80
 - c. 78
 - d. 75.5
- 69** A survey combines two groups: 20 people with average age 30, and 30 people with average age 40. Find the combined average age.
- a. 70
 - b. 35
 - c. 36
 - d. 34
- 70** A class of 8 students has an average score of 65. A 9th student joins, changing the new average to 68. Find the 9th student's score.
- a. 71
 - b. 65
 - c. 88
 - d. 92
- 71** Two classes combine: Class A (10 students, mean 60) and Class B (n students, mean 90), giving a combined mean of 78 for all students. Find n .
- a. 12
 - b. 15
 - c. 10
 - d. 20

MCQ — Comparing data sets

- 72 Data set A: 12, 14, 16, 18, 20. Data set B: 8, 15, 17, 19, 21. Both have the same mean. Which has the greater range?
- a. Cannot be determined
 - b. They are equal
 - c. Data set A
 - d. Data set B
- 73 Team X scored 18, 22, 25, 23 points in 4 games. Team Y scored 15, 15, 32, 26. Both teams have the same mean. Which team is more consistent (smaller range)?
- a. Team X
 - b. Team Y
 - c. Cannot be determined
 - d. They are equally consistent
- 74 Class 1 has 16 students with mean score 70; Class 2 has 24 students with mean score 80. Find the combined mean of both classes.
- a. 150
 - b. 75
 - c. 76
 - d. 77

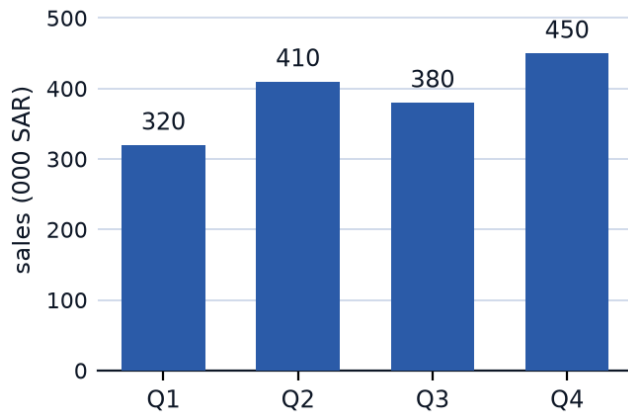
- 75 Data set A: 5, 10, 15, 20, 25. Data set B: 13, 14, 15, 16, 17. Both have mean 15. Which has the smaller range, and what is it?
- a. Data set B; 20
 - b. Data set B; 4
 - c. Data set A; 4
 - d. Data set A; 20
- 76 Two data sets both have median 20. Set P: 10, 15, 20, 25, 30. Set Q: 18, 19, 20, 21, 22. Which set is more spread out?
- a. Set P
 - b. Cannot be determined
 - c. Set Q
 - d. They are equally spread out
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MCQ — Word problems

- 77 A teacher records the number of siblings for 12 students: 2, 1, 0, 2, 3, 1, 2, 0, 1, 2, 3, 1. Find the mean, to two decimal places.
- a. 1.33
 - b. 1.60
 - c. 1.50
 - d. 2.00

- 78 A company's weekly sales (in thousands of riyals) over 6 weeks: 45, 60, 38, 72, 55, 50. Find the mean weekly sales.
- a. 55.00
 - b. 320
 - c. 50.00
 - d. 53.33
- 79 A jar contains 6 red, 4 blue and 2 yellow candies. A candy is picked and eaten, then another is picked. Find $P(\text{both red})$.
- a. $\frac{6}{12}$
 - b. $\frac{5}{22}$
 - c. $\frac{1}{4}$
 - d. $\frac{3}{11}$
- 80 In a survey of 300 students, 180 prefer math and the rest prefer science. Find the probability, as a percent, that a randomly chosen student prefers science.
- a. 120%
 - b. 40%
 - c. 18%
 - d. 60%

- 81 The bar chart shows a company's quarterly sales, in thousands of riyals. Find the total annual sales, and the mean quarterly sales.



- a. 1,560; 450
- b. 1,560; 390
- c. 1,560; 400
- d. 1,200; 390
- 82 A survey of 150 shoppers found 90 prefer online shopping and the rest prefer in-store. Find the probability, as a fraction in simplest form, that a randomly chosen shopper prefers in-store.
- a. $\frac{2}{5}$
- b. $\frac{90}{150}$
- c. $\frac{3}{5}$
- d. $\frac{3}{50}$
- 83 A basketball player makes 18 out of 25 free throws attempted. Find the probability, as a decimal, that the next free throw is missed (based on this rate).
- a. 0.28
- b. 0.18
- c. 0.25
- d. 0.72